

Data Lineage

Происхождение данных

LMS API → Raw S3 (students, courses, grades) → Bronze → Silver → Gold (student_features)

CSV → Raw S3 (schedule, rooms, teachers) → Bronze → Silver

Kafka Events → ClickHouse (people_count) — агрегация 5-мин окон

Зависимости между датасетами

Входной датасет	Трансформация	Выходной датасет
Raw students	Bronze	Bronze students
Raw grades	Bronze	Bronze grades
Bronze students	Silver (очистка)	Silver students
Bronze grades	Silver (фильтрация)	Silver grades
Silver students + grades	Gold (агрегация)	Gold student_features
Kafka uni_events	Aggregator	ClickHouse people_count

Сквозная трассировка

Событие: student.entered.classroom

Kafka → Aggregator (5-мин окна) → ClickHouse → Cube.js → Streamlit

Оценка студента:

LMS API → Airflow DAG → Raw S3 → Bronze S3 → Silver S3 → Gold S3 → ClickHouse
→ Cube.js → Streamlit